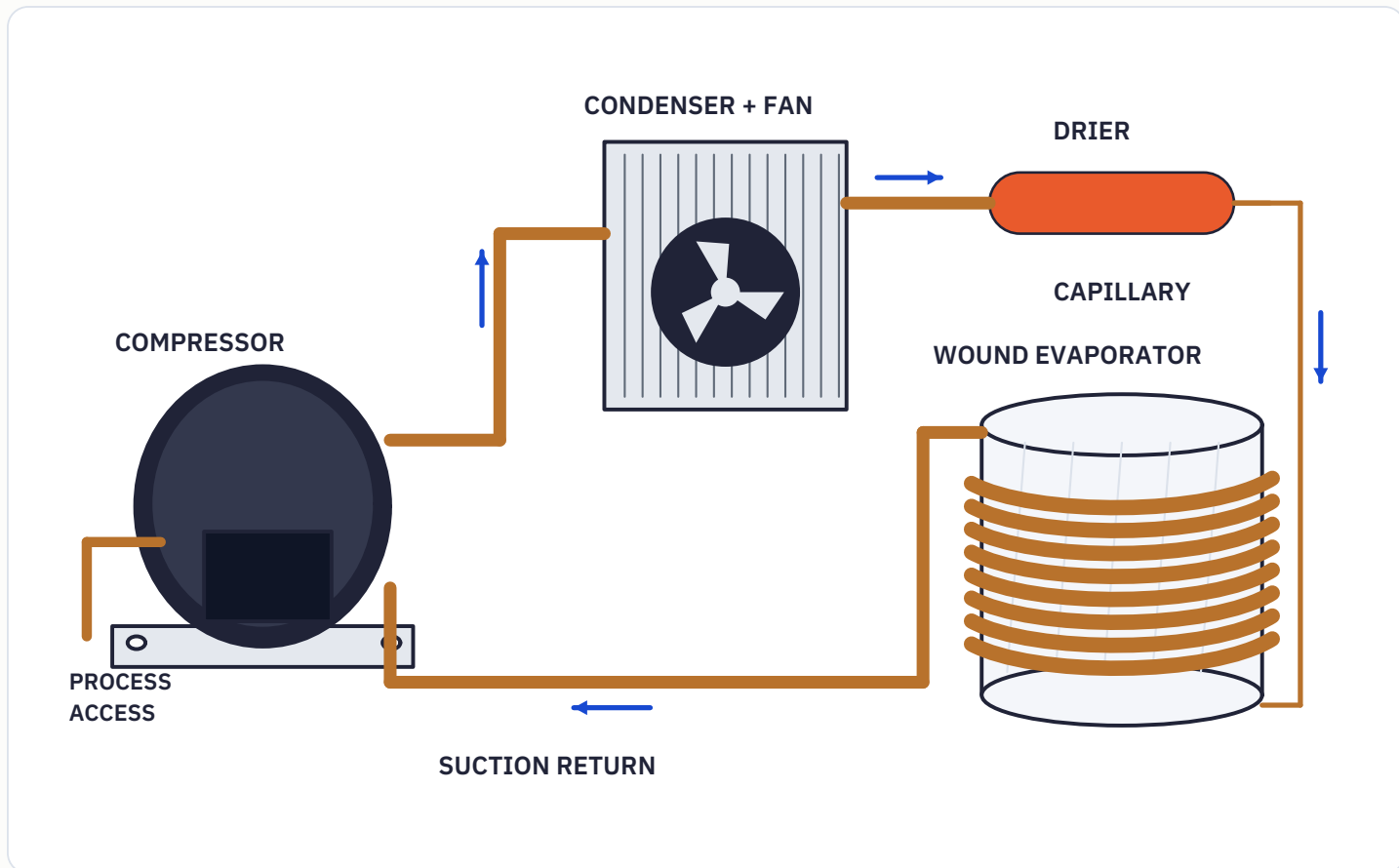


Build the cold loop

Wind the evaporator, transfer the donor circuit, and prove the installed refrigeration assembly.



1 Follow the physical sequence

Coil work: pages 4-8. Donor prep and circuit service: 2-3 and 9-20. Mounting and commissioning: 21-28.

2 Read the pictures

Coral marks the current work. Copper is refrigerant tube. Blue marks a tool, gauge or movement. Pictures are schematic; printed dimensions govern.

3 Keep the scope honest

Dry coil and mount work have bench steps. Opening or heating R600a requires the qualified service setup described at the actual operation.

CHECK

A cured cold core alone does not qualify a refrigerant circuit or a finished charge.

Read the donor's label

Identify the exact appliance and compressor before choosing the service route or supplying power.

APPLIANCE RATING LABEL

MODEL

Read exact model

REFRIGERANT

R600a

CHARGE

Copy actual grams

VOLTAGE / Hz

Copy actual supply

REFERENCE BASELINES

Unit A / HZB-12/Q **15 g**

Unit B / EFIC117-SS **23 g**

The donor's label governs.

Cast-stamps 48.5-2 / 45
are not charge masses.

1 Photograph the real labels

Record appliance model, refrigerant, factory charge, compressor voltage, frequency and current ratings. Keep each donor's data with its parts.

2 Verify R600a

Use the actual label. A different refrigerant changes handling and circuit suitability. The referenced 15 g and 23 g values are donor baselines.

3 Mark the assembly

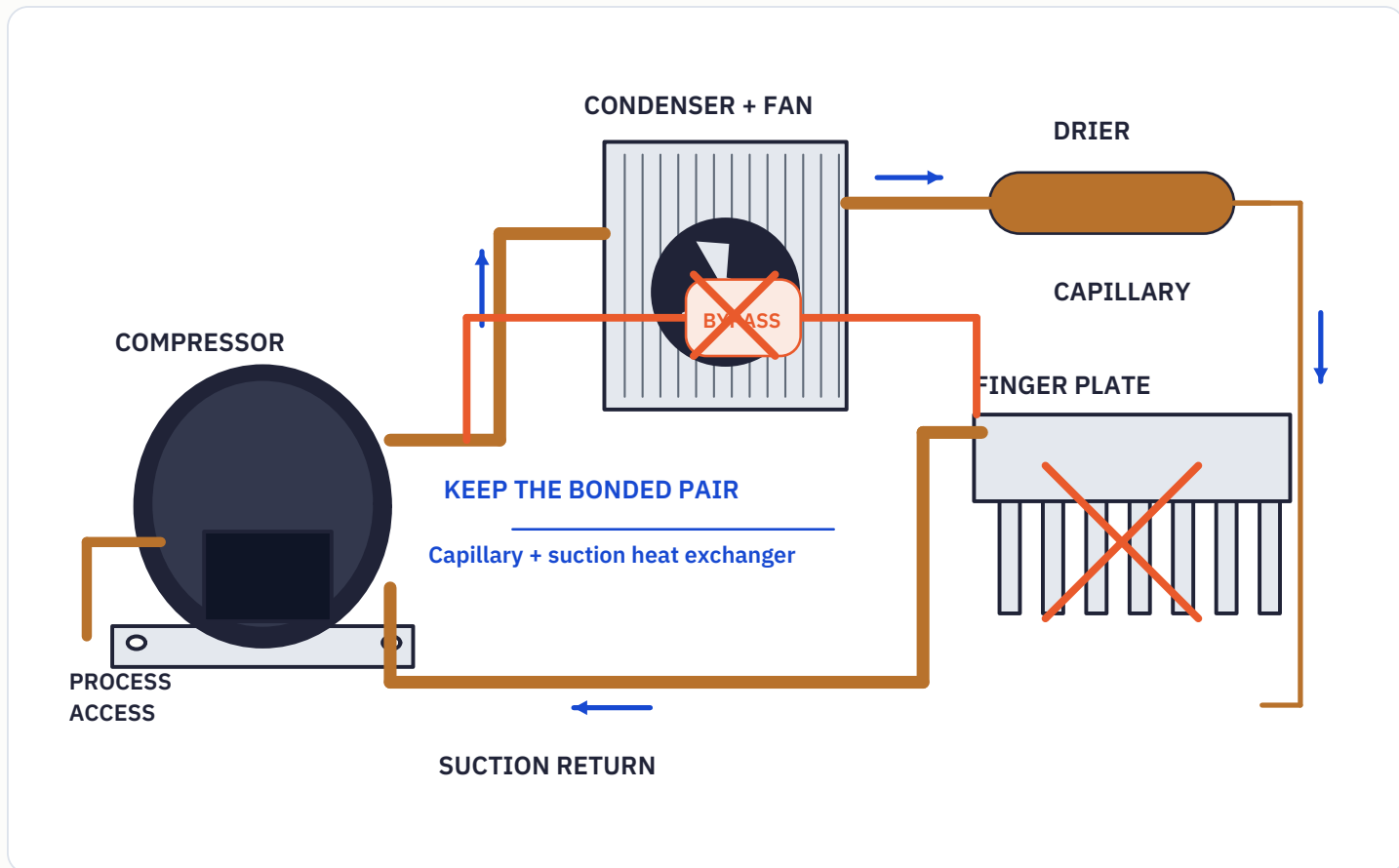
Tag Unit A or Unit B and its compressor. Do not infer charge from the cast-stamps or borrow Unit A's electrical lead colors for Unit B.

DONE WHEN

The labels and supply ratings are recorded; the finished-machine charge remains separate.

Trace what stays

Walk every tube while the donor is still intact. Mark the cuts and closure plan before dismantling.



1 Keep the useful assembly

Compressor, condenser/fan, original terminal cover, grommets and the bonded capillary/suction pair stay together.

2 Remove the ice-making path

The finger plate and hot-gas bypass serve ice harvest. Their removal must leave a continuous normal circuit with no bypass or open tee.

3 Plan the drier replacement

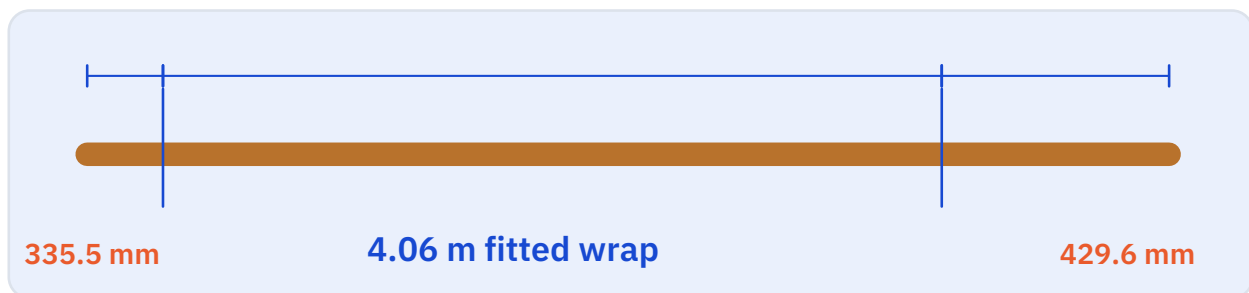
The loop-opening session consumes a compatible new drier. Record its capillary connection and preserve metering length or specify a recalculation.

HOLD

Unit A's topology is recorded. Trace Unit B at teardown; its exact branch closure is open.

Reserve both coil tails

Cut the dry coil stock with a tubing cutter, with enough copper for the wrap and each separate routed tail.

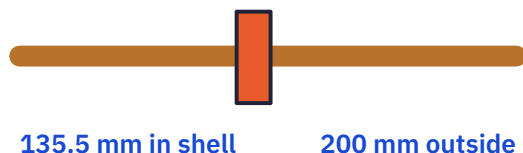


Inlet allowance

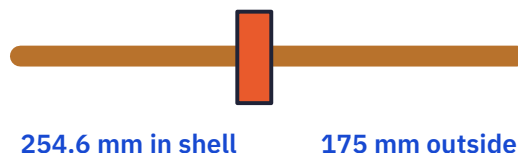
Outlet allowance

Total cut / about 4.85 m (15.92 ft)

INLET / plug face is the datum



OUTLET / plug face is the datum



length picture is schematic

1 Use the specified stock

GOORY ACR copper, 1/4 in OD x 0.031 in wall. Keep its bore clean and capped. Do not saw refrigerant tubing.

2 Mark the wrap and tails

Current nominal stock is about 4.85 m total: 4.06 m fitted wrap plus 335.5 mm inlet and 429.6 mm outlet allowances.

3 Keep the allowances intact

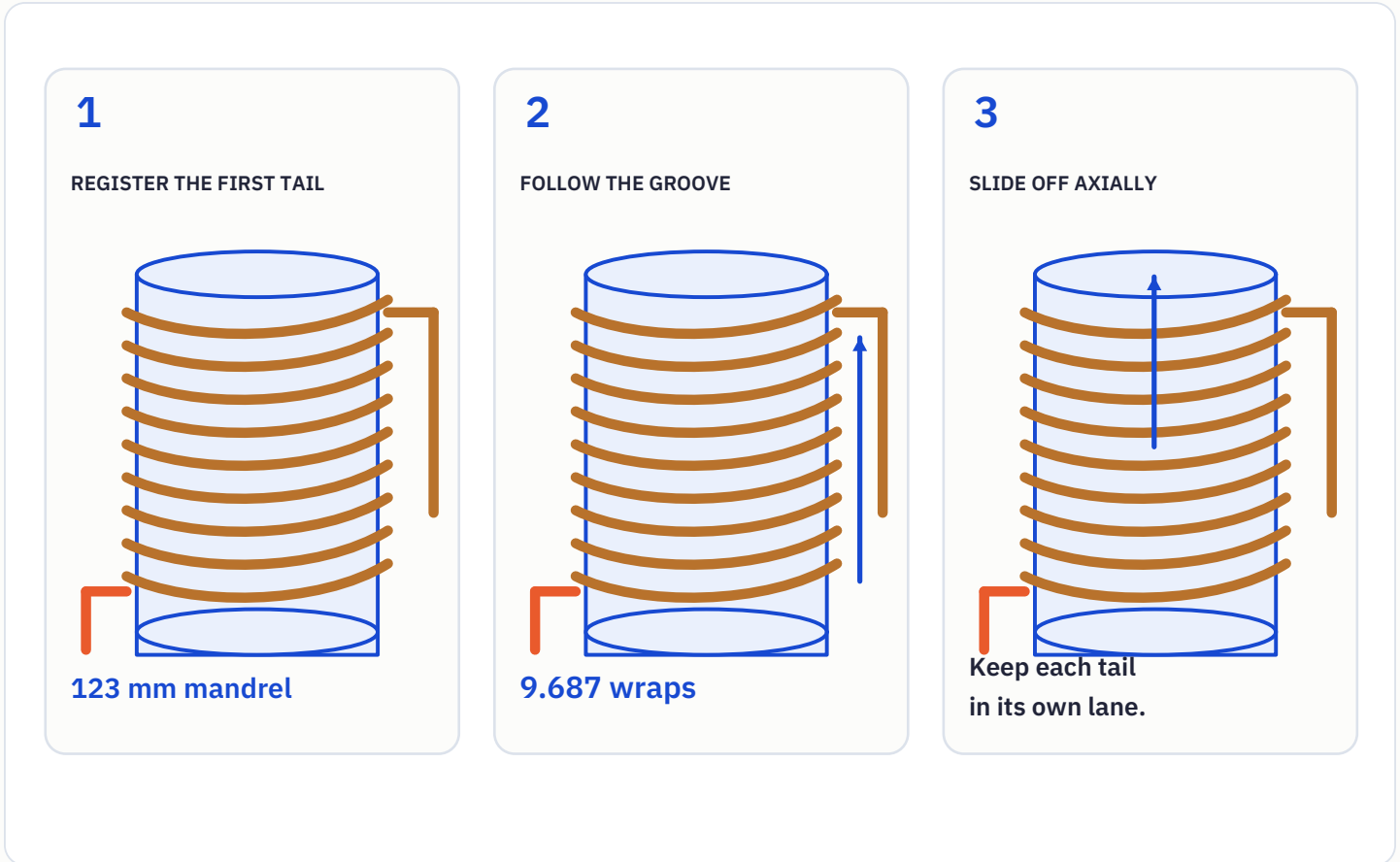
Each allowance includes the in-shell route and protrusion. The longer outlet route still leaves the shorter outside working length.

CHECK

The two tails are labeled and the ends are round, square, burr-free and protected from dirt.

Wind the copper coil

Use the printed mandrel groove to place a single layer of tubing without flattening the bore.



1 Start on the correct tail

Reserve the inlet allowance, register the first wrap in the groove, and keep the tail on its assigned side.

2 Follow the helix

Wind 9.687 wraps at 12.33 mm pitch over 119.4 mm. The mandrel is 123 mm OD. Keep steady contact without crushing or kinking the tube.

3 Remove without bending a tail

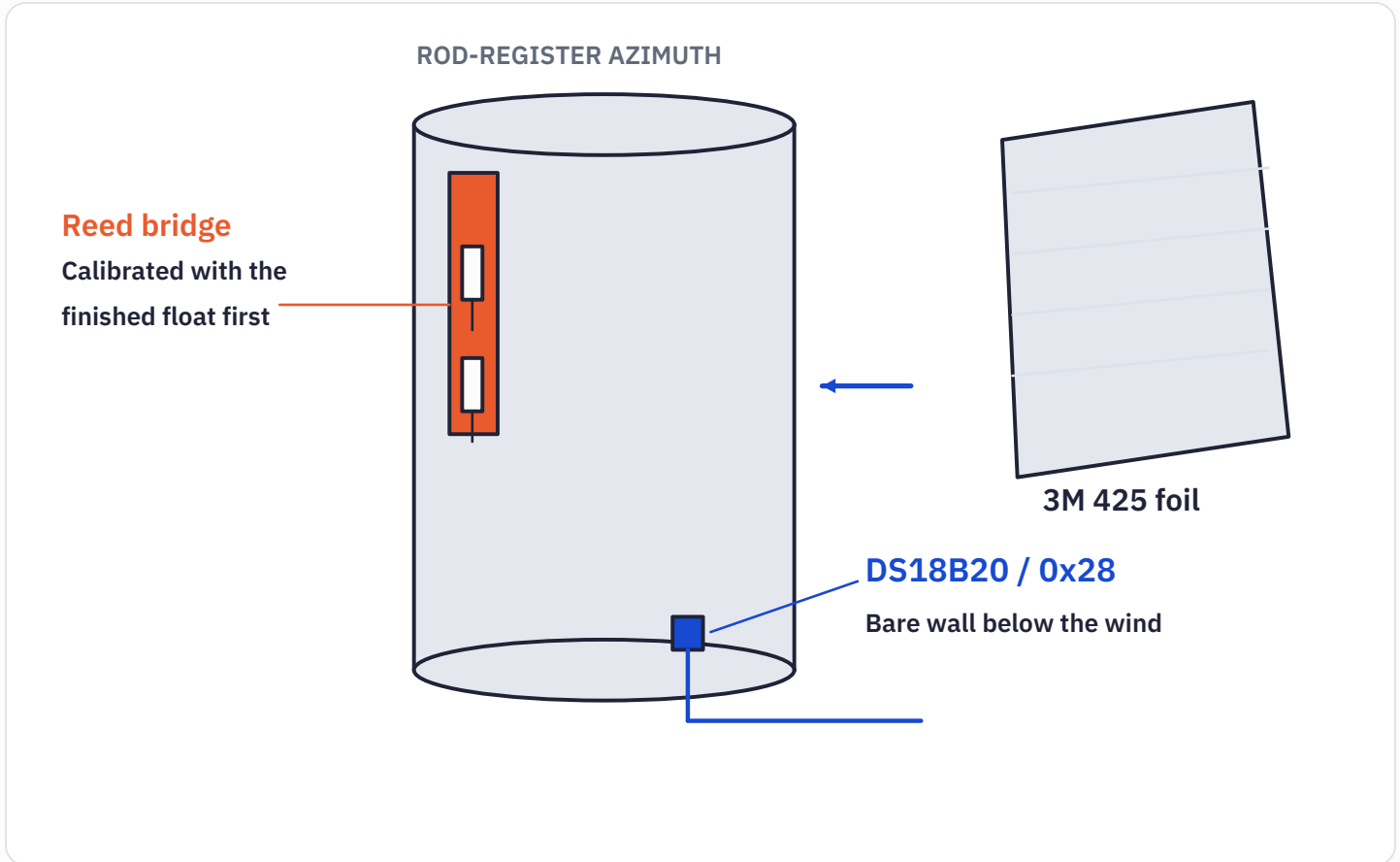
Slide the coil axially off the mandrel. Inspect every turn and both exits; a flattened or sharply creased tube is not usable.

DONE WHEN

The free coil holds one layer and the inlet/outlet tails leave on opposite assigned lanes.

Dress the bare wall

Install and prove everything that needs direct steel contact before foil, coil or foam covers it.



1 Bond the wall probe

Tape the DS18B20, family 0x28, flat against bare carbonator steel in the band below the wind. Insulate its leads and route them upward.

2 Set the reed bridge

Use the rod-register azimuth. Calibrate the finished float and both directional liquid crossings before fixing the final bridge position.

3 Apply the foil skin

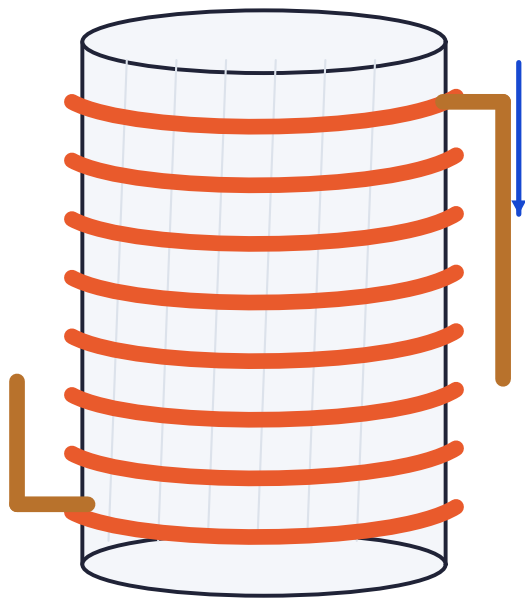
Apply continuous 3M 425 foil between steel and coil. Preserve the proven sensor positions, lead insulation and the bridge's crossing envelope.

DONE WHEN

Probe and reed checks are recorded while every contact point is still reachable.

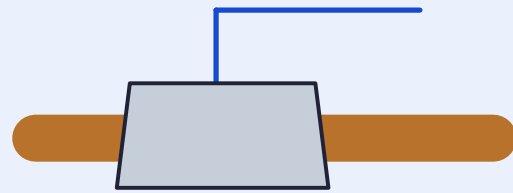
Transfer and bond the coil

Slip the wound coil over the foil-dressed 127 mm carbonator, then place the freeze-protect probe.



127 mm carbonator

SUCTION-END PROBE



DS18S20 / 0x10

Flat on copper at the
high / outlet end

Secure before foam hides it.

1 Slide the coil into place

The wound coil is undersize to hold against the tank. Ease it on without kinking a tail, dislodging a reed or abrading insulated leads.

2 Add the coil probe

Tape the DS18S20, family 0x10, flat against suction-end copper at the high/outlet end. The wall probe stays on steel below the wind.

3 Secure and prove

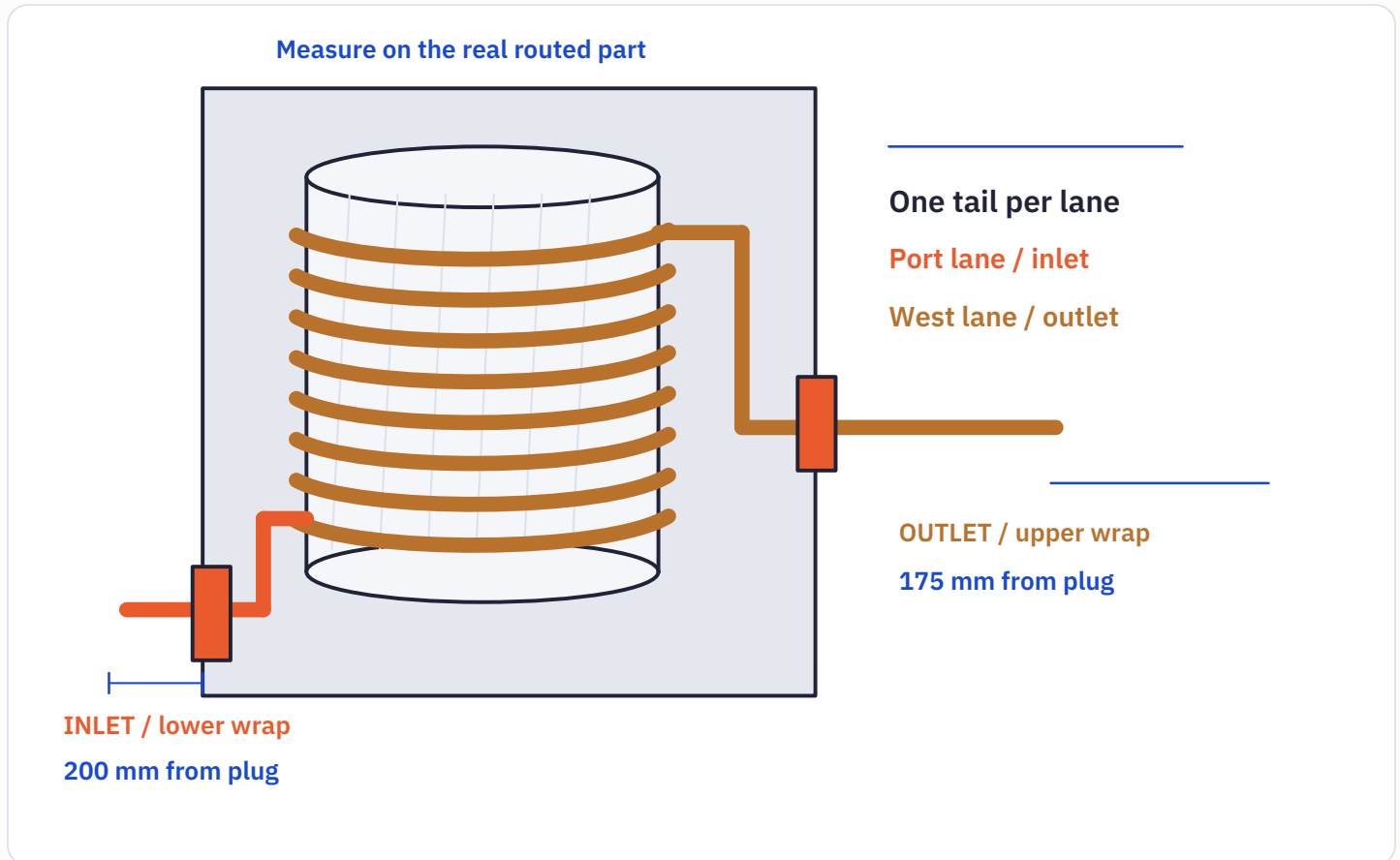
Bond the thermal contact with foil. Prove both probe families and lead continuity before the mold guide's pre-pour check hides the surfaces.

CHECK

Two different probe families: 0x28 on steel; 0x10 on copper. Neither probe floats in foam.

Route the two exits

Lower the dressed carbonator into the open shell with one refrigerant tail in each assigned lane.



1 Clock by the tails

The lower/inlet tail takes the port lane. The upper/outlet tail takes the west lane. Avoid torsion where either tail leaves the wrap.

2 Measure from each plug face

Leave 200 mm inlet and 175 mm outlet outside the plug. Measure the physical routed assembly; do not cut away the in-shell part of an allowance.

3 Handoff to the mold guide

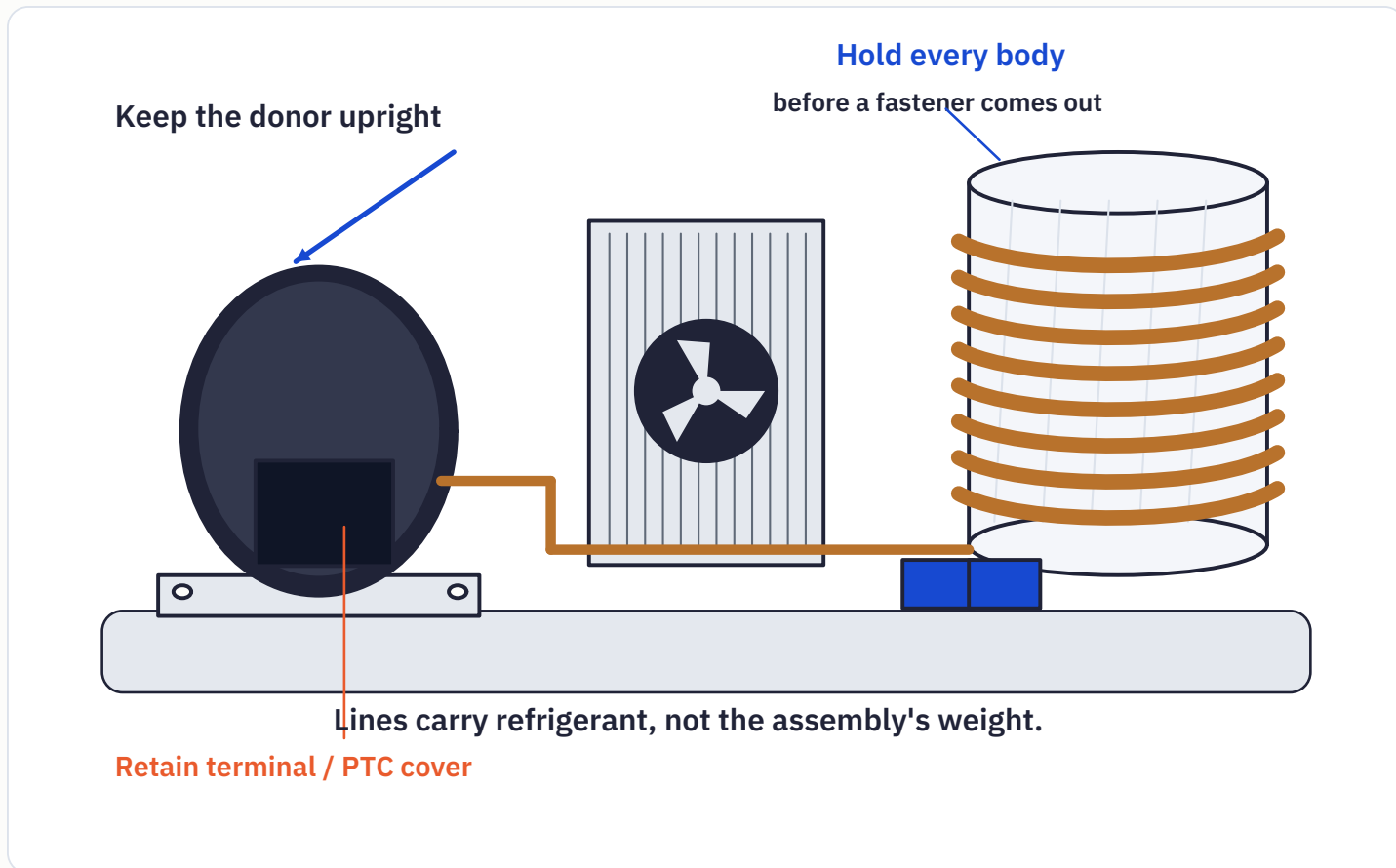
The copper plugs close their elongated wall slots, with sensor leads protected. Follow the mold guide for pre-pour inspection and cured closure.

DONE WHEN

Both protrusions are recorded and the cured core can be supported with no load on the tube exits.

Support before teardown

Remove only the donor casing and supports that prevent access, while keeping the sealed circuit undamaged.



1 Unplug and identify the leads

Keep mains isolated. Photograph the factory-external compressor and fan connections. Retain the terminal/PTC cover and its original retention.

2 Support each body

Keep the compressor upright. Support the condenser, fan and evaporator before undoing their fasteners. Copper tubing must not hold their weight.

3 Preserve the useful details

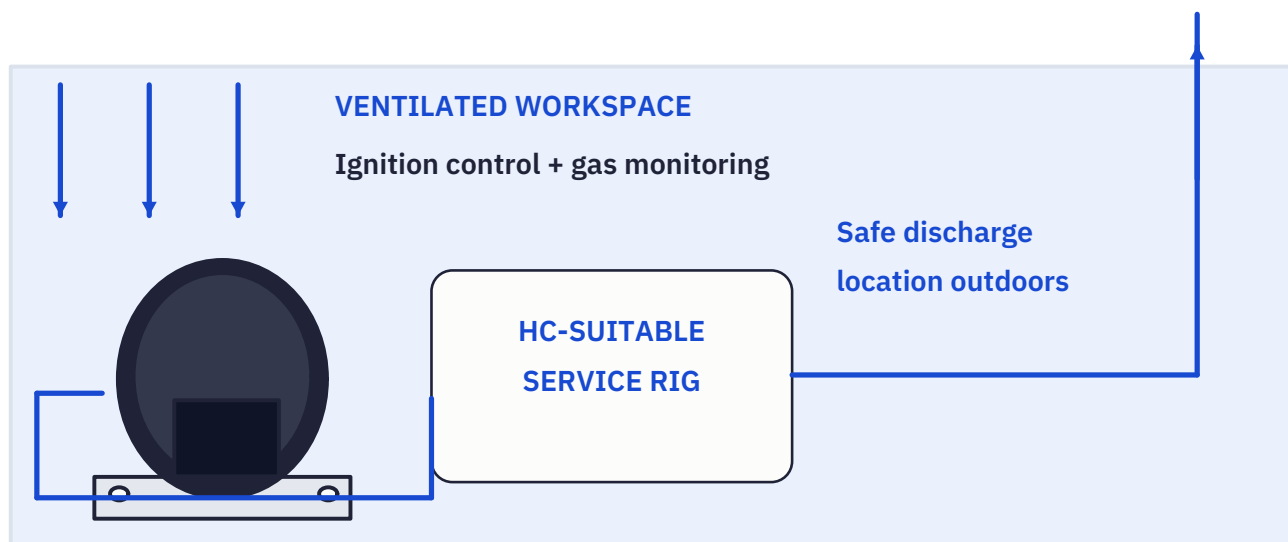
Retain grommets and mounting hardware for inspection. Protect condenser fins, the bonded capillary pair and the process stub. Keep Unit B lead identity open until recorded.

CHECK

A missing, cracked, altered or loose terminal cover fails the assembly before power is applied.

Make the HC bench ready

Complete the service setup before piercing, cutting, pressure testing, evacuation or charging.



Equipment approval

Exact model + R600a use

Written service setup

Removal / purge / leak / charge

1 Assign the service work

A technician trained for R600a chooses the handling route and approves the exact equipment, hoses, connections, ventilation and ignition control.

2 Stage the complete session

Have the replacement drier, joint arrangement, final process closure, dry-nitrogen controls, leak-test method, micron gauge and charge procedure ready.

3 Verify the real equipment

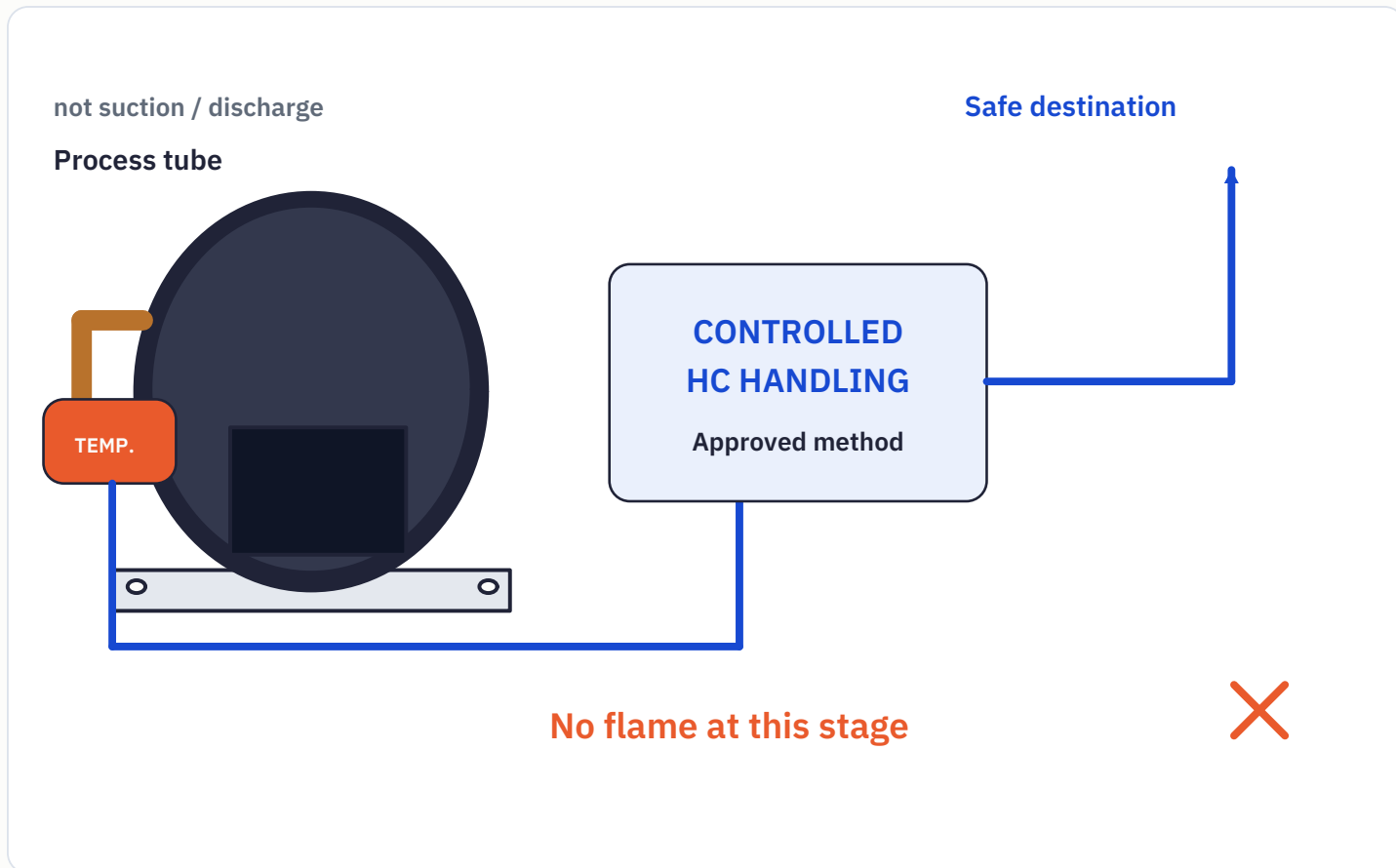
An ultimate vacuum figure and a 1/4 in SAE fitting do not establish hydrocarbon suitability. The owned Orion gear has no recorded exact-model HC approval.

HOLD

Opening waits for this setup. The technician's approved method supplies the missing circuit limits.

Remove the donor charge

Use the factory process tube as controlled service access under the approved hydrocarbon-handling method.



1 Fit temporary access correctly

A BPV31 may be fitted if its adapter and instructions match the real tube. Connect the handling rig before piercing or opening.

2 Use the approved destination

The technician records donor end-use, handling route and discharge/collection arrangement. Keep ignition sources outside the defined service area.

3 Prove the clearing condition

Depressurize under the service method. Residual fuel can leave compressor oil; gauge zero and absence of smell alone do not authorize a cut or flame.

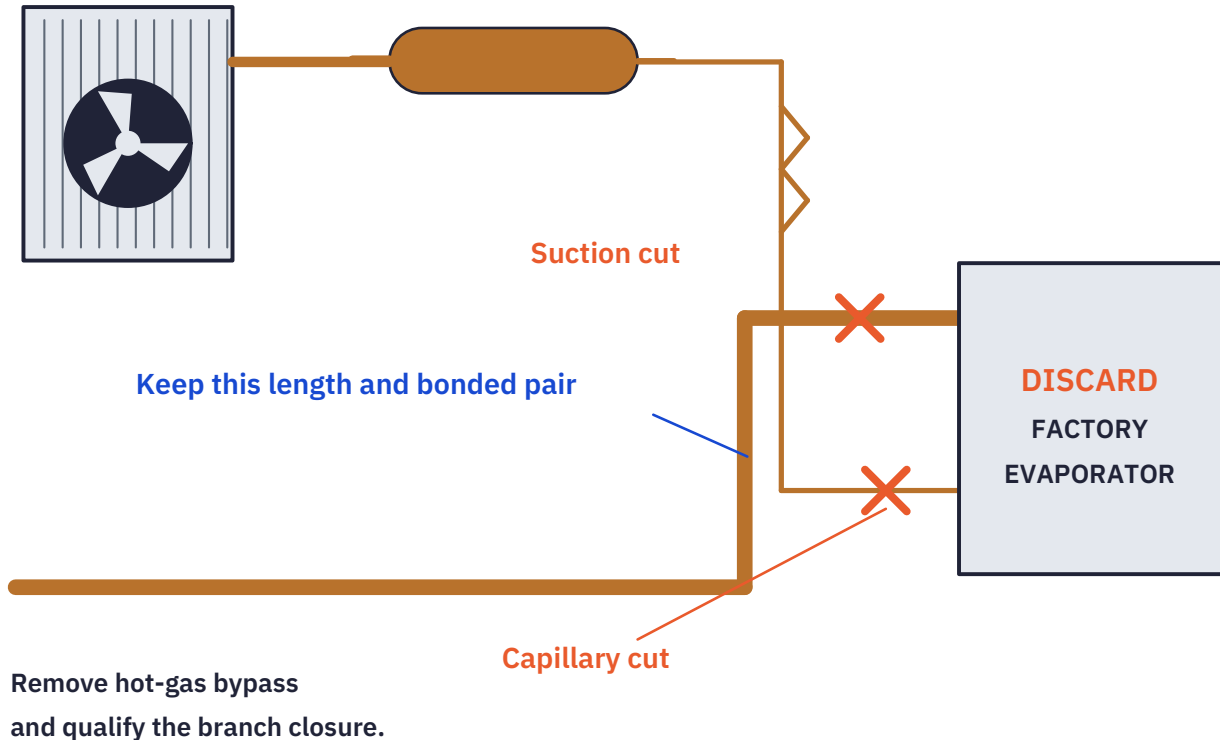
CHECK

EPA's exemption depends on end use. The donor's status does not authorize venting a rebuilt unit.

Cut at the evaporator

Remove the ice-making evaporator while preserving the donor's metering restriction and bonded heat exchanger.

Trace your donor; cut labels are functional.



1 Mark the two functional cuts

Suction side: near the discarded evaporator outlet. Capillary side: at its evaporator-inlet end. Trace the actual donor before cutting.

2 Use the dedicated cutters

Tubing cutter for normal copper; Mastercool capillary cutter for the hair-bore tube. Clean the outside first and keep every burr or grit particle out.

3 Close the bypass correctly

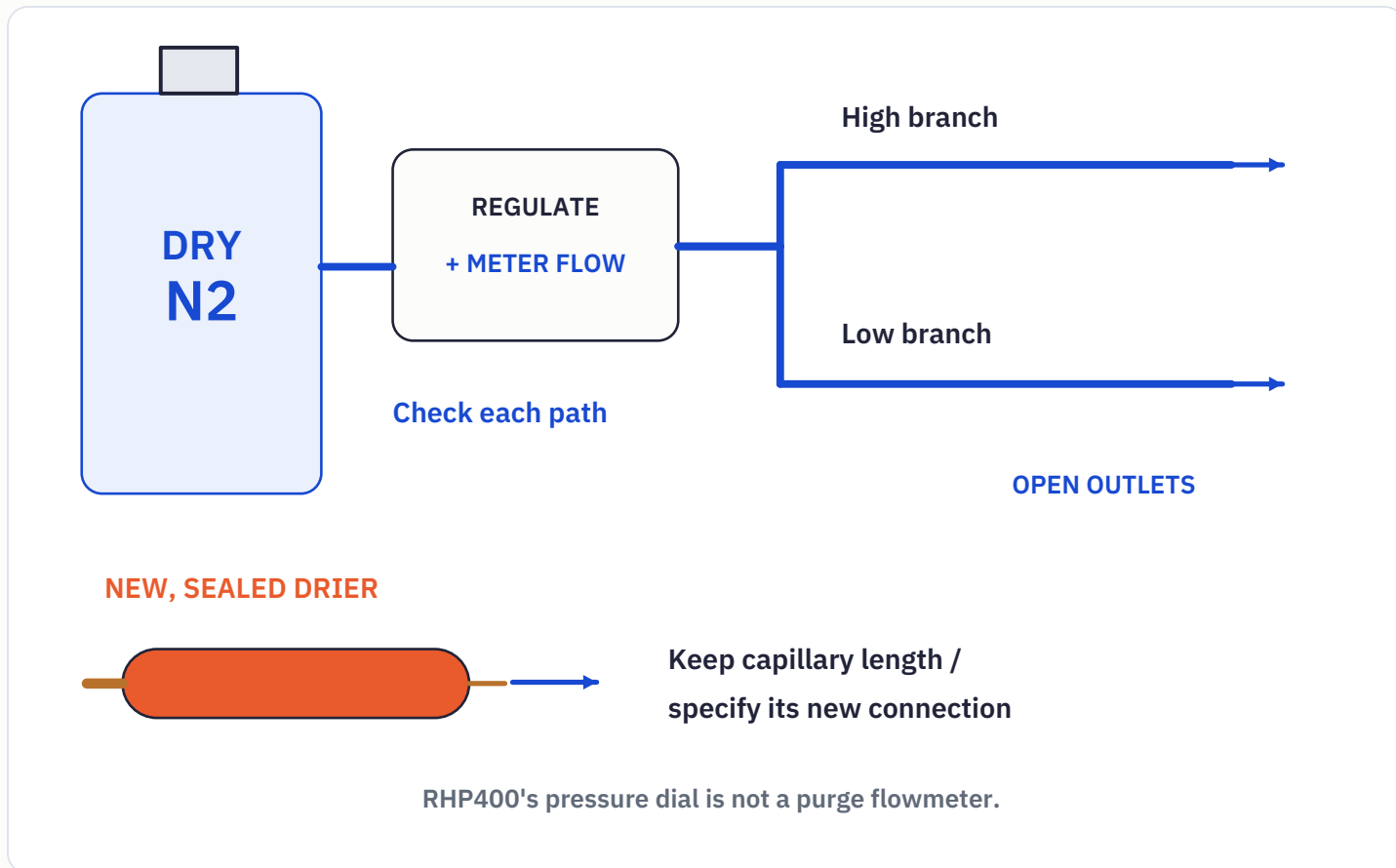
Remove the harvest solenoid/branch under the donor-specific closure plan. Keep the capillary helix and bonded suction pair without shortening or pulling them apart.

HOLD

No improvised tee cap or guessed metering length: qualify the traced branch closure and capillary plan.

Clear, then replace the drier

Use the technician's dry-nitrogen sequence to clear each circuit branch and protect every braze internally.



1 Clear high and low paths

Verify the actual gas path and open outlet for each branch. A pressure dial does not prove protective flow through the capillary or compressor.

2 Fit a fresh compatible drier

Cut the used drier out without heating it. Keep the selected replacement sealed until assembly; specify its connection to the retained capillary.

3 Protect the open loop

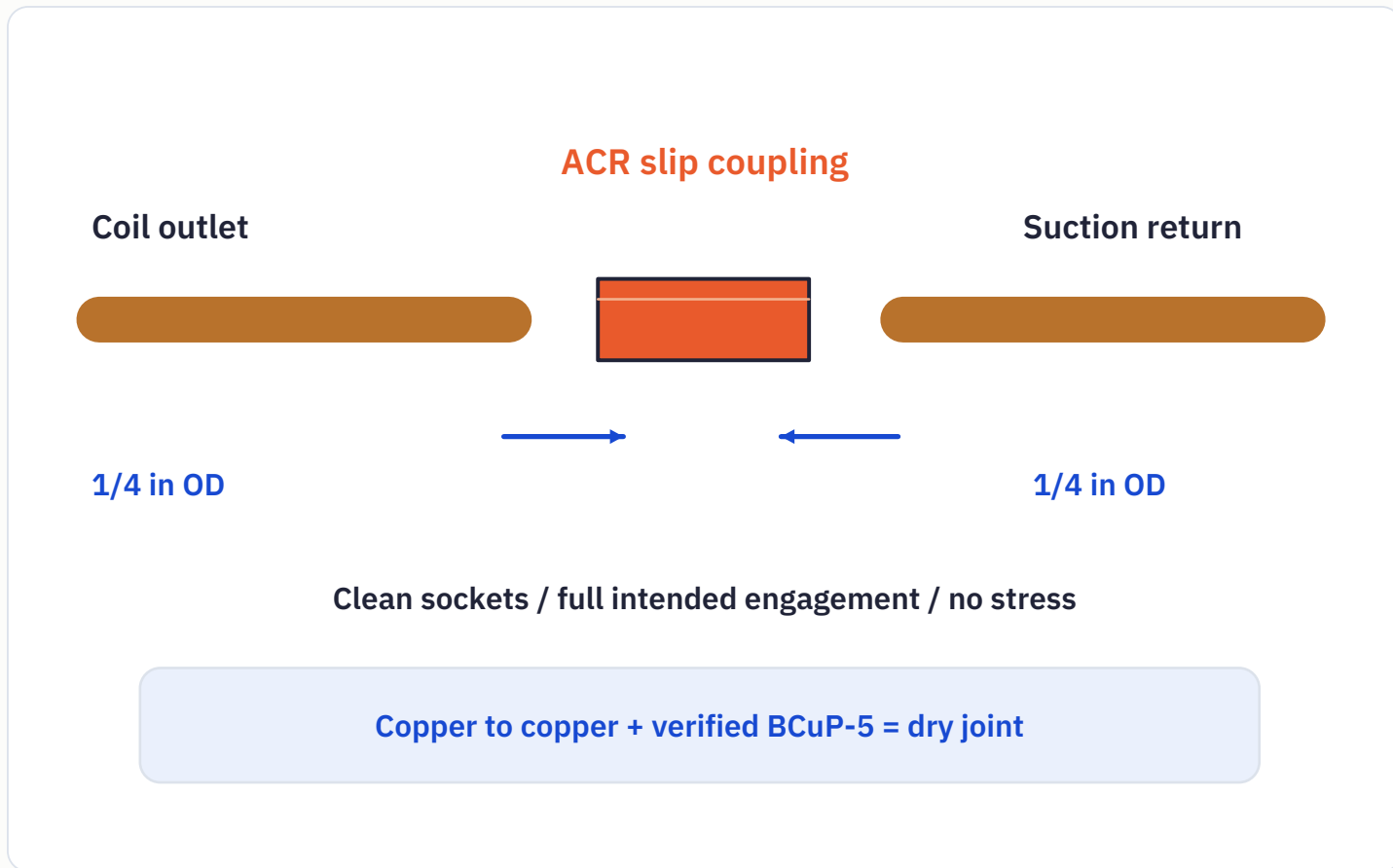
Continue approved dry protective flow through a joint while brazing. Close and protect any interrupted work against moisture. Keep an unobstructed outlet.

HOLD

Nitrogen pressure/flow and drier compatibility are specified by the service plan, not the picture.

Fit the suction coupling

Join the upper coil outlet to the retained factory suction line with an ACR-grade 1/4 in slip coupling.



1 Prepare the dry fit

Clean both tube ODs and fitting sockets. Preserve a clean bore. Confirm the coupling takes the actual tube sizes and intended engagement.

2 Align without spring load

Support the core and donor assembly at their installed relationship. Each tube enters straight; the joint must not pull the plug or hang a component.

3 Stage for brazing

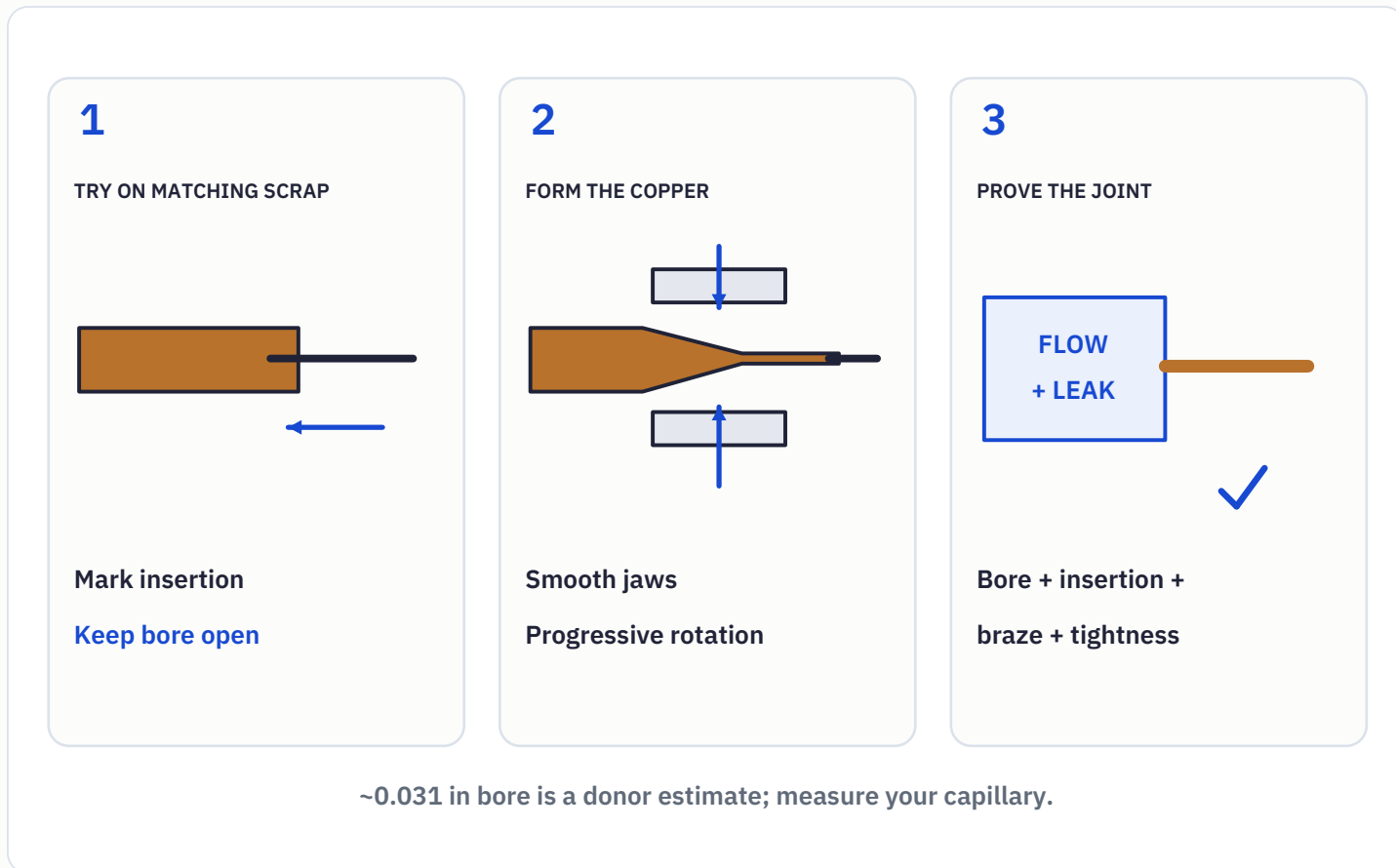
For verified copper-to-copper joints, BCuP-5 is self-fluxing. Keep flux out of the sealed loop. Use the qualified purge and plug-protection sequence on page 16.

DONE WHEN

The intended engagement is visible/marked, the joint is unstressed and the bore stays clean.

Prove the capillary joint

The proposed pinch-swage reduces the coil-inlet tube around the donor capillary. Qualify a sample before the core.



1 Measure and try the real pair

The cited ~0.031 in capillary bore is an estimate. Use matching stock. Establish insertion, forming and braze details on a representative sample.

2 Form without closing the bore

The proposed tool is smooth parallel-jaw Knipex pliers. Progressive rotation collapses the larger tube; the 60-degree suggestion is a trial, not an accepted recipe.

3 Record what the sample proves

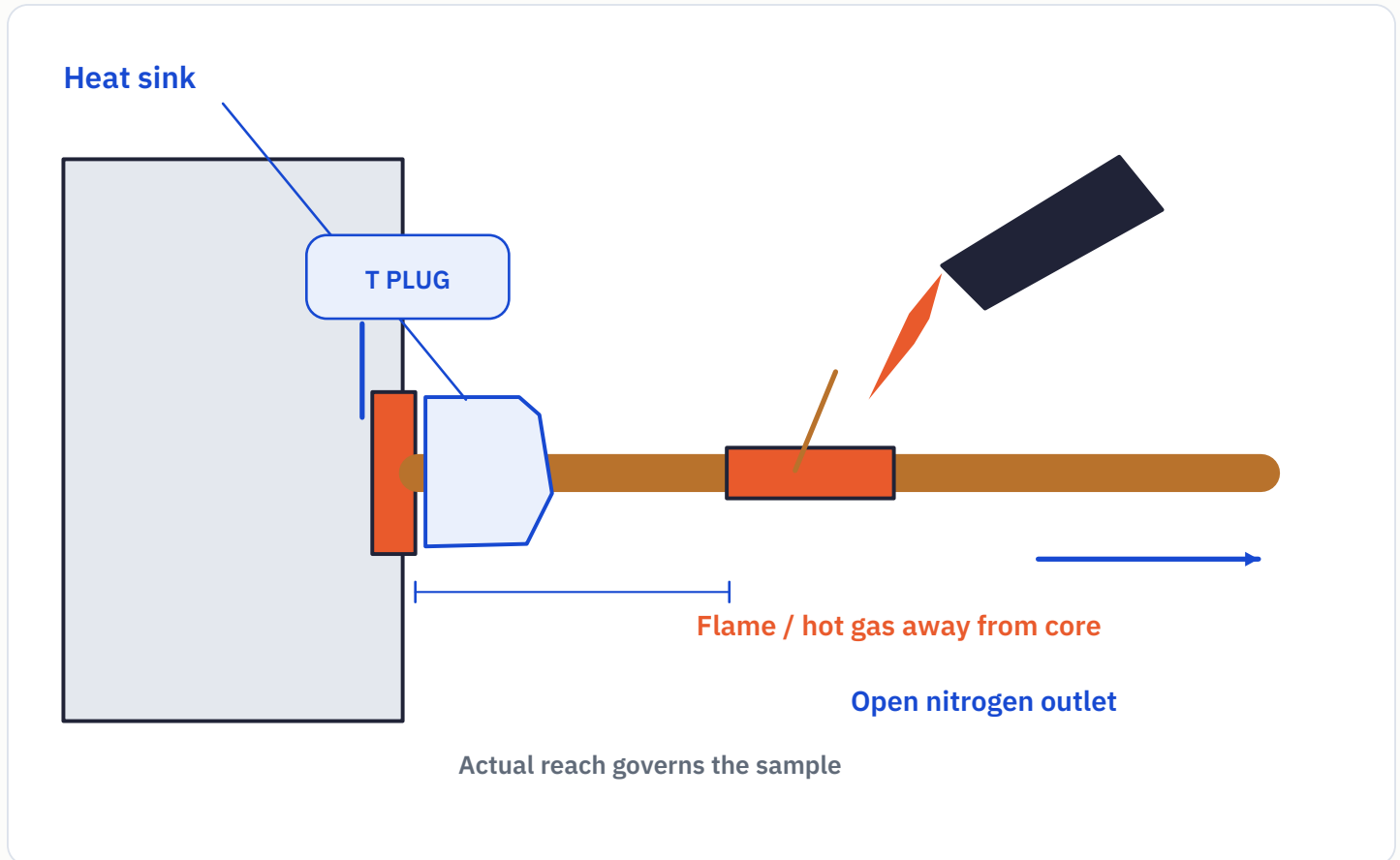
Prove the capillary remains open, the required insertion is retained, the braze is sound and the joint meets the specified leak test. Apply only the accepted sequence to the core.

HOLD

The metering-joint qualification is open. A closed-looking swage does not prove a working capillary.

Protect the plug, then braze

Use the qualified heat sink, flame shield and temperature limit at both installed-core joints.



1 Qualify the thermal stack

Use the same copper reach, PETG plug and foam as the installed joint. Measure plug-interface temperature through heating and cooling; justify its allowed limit.

2 Make the controlled braze

With residual-fuel clearing complete and protective nitrogen flowing, use the accepted joint sequence. Direct flame and hot gas away from the core.

3 Stop and inspect

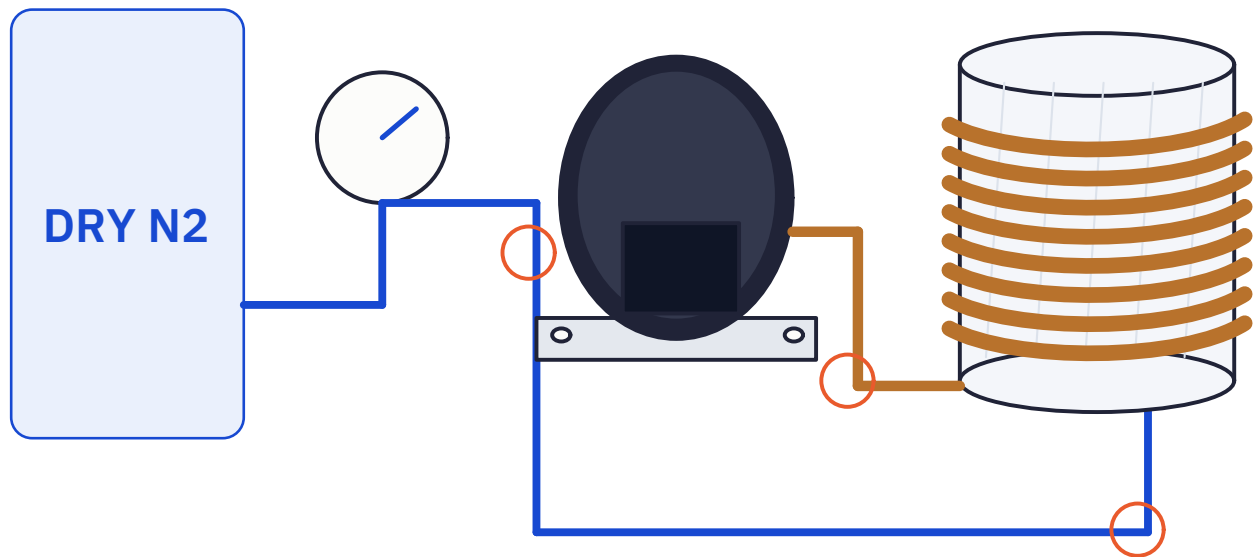
Stop at the qualified temperature/time limit or a failing heat-sink contact. Let the joint cool, then inspect the plug and foam boundary before the second braze.

HOLD

A wet rag's boiling-point behavior is above nominal PETG T_g (~80 C). It does not qualify this stack.

Test the closed circuit

Pressure leak-test after all joints are complete and cool, before evacuation and the first R600a charge.



TEST PRESSURE + TIME + DECAY LIMIT

Specified from the weakest rated circuit component

No pressure figure is inferred from the water-side hydro-test.

1 Use the written test limits

Dry nitrogen, regulated and pressure-protected. The technician specifies pressure, time, temperature correction and decay acceptance from the weakest rated component.

2 Inspect every new connection

Include both coil joins, replacement drier joins, hot-gas branch closure and the planned final service/process connection. Use the qualified leak-test method.

3 Record and depressurize

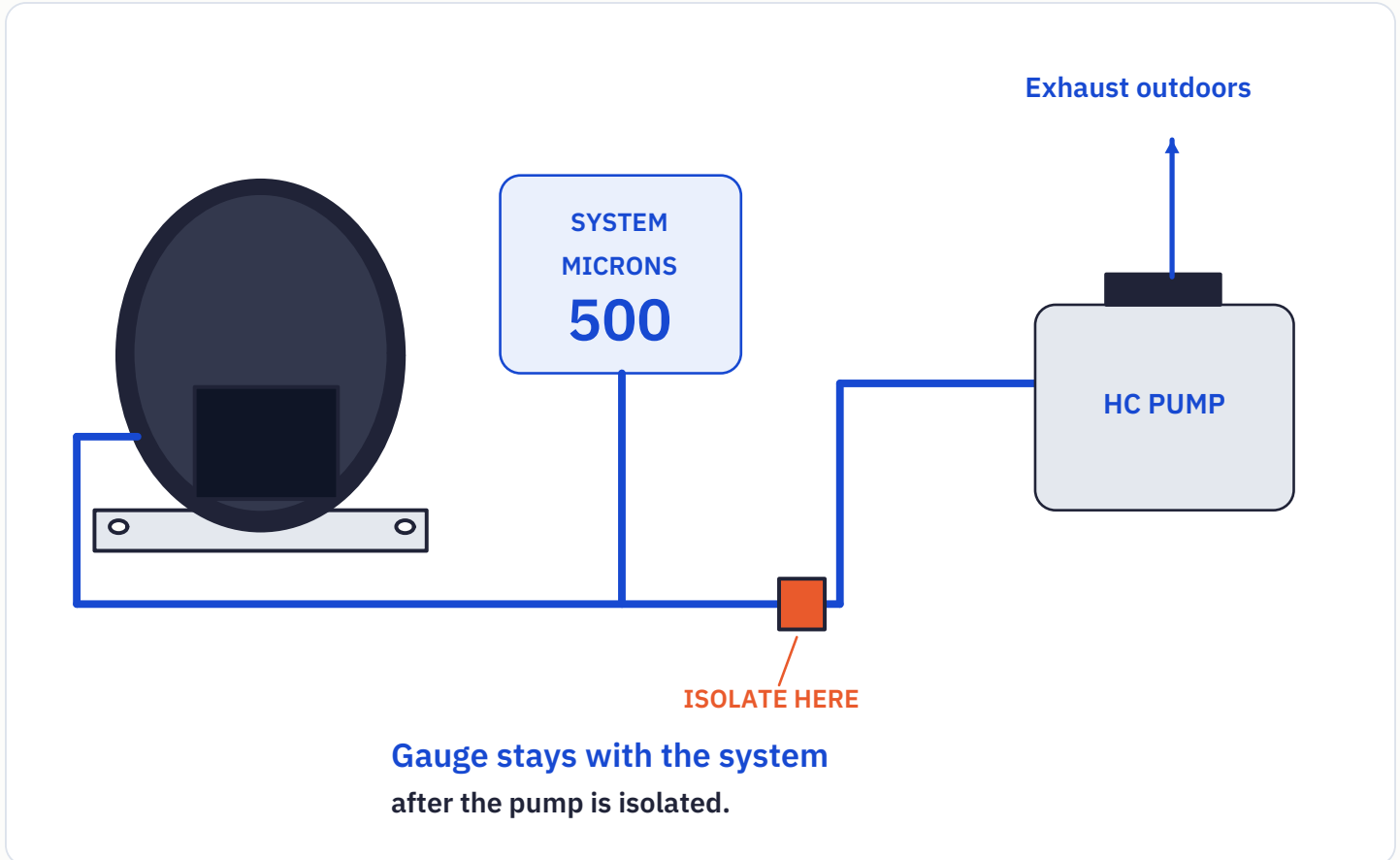
Log the gauges, starting/ending pressure, elapsed time and all inspected sites. Release test gas under the service method before connecting the vacuum rig.

HOLD

Circuit pressure-test limits are open. The stainless water-vessel hydro-test does not supply them.

Measure the vacuum

Read absolute pressure at the circuit with suitable HC evacuation equipment; keep that gauge on the isolated system.



1 Place the system gauge

An absolute micron gauge reads the circuit, not the isolated pump. Keep the compressor off and route pump exhaust to the approved outdoor location.

2 Reach the project target

The current target is 500 microns or less, with at least 15 minutes while pumping. Record the actual pressure trace; a compound manifold dial cannot prove it.

3 Isolate and observe

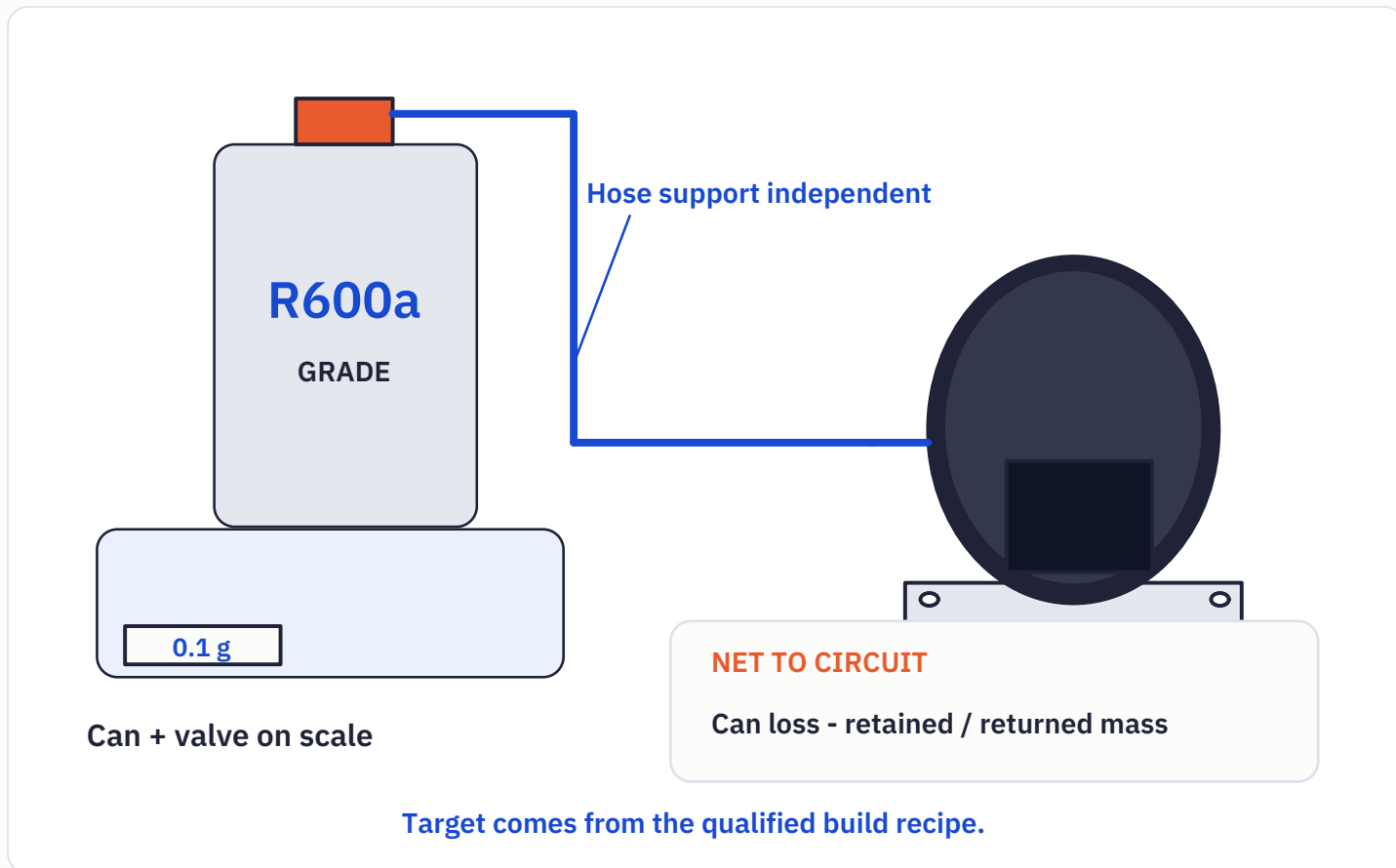
Isolate the pump and record the next 15 minutes at the system gauge. Diagnose a rapid rise. Some rebound is possible; use the technician's written decay limit.

HOLD

The isolated-pressure acceptance and exact equipment approval must be settled before charging.

Meter the qualified charge

Use the finished-machine charge recipe and account for the refrigerant that remains outside the circuit.



1 Start from an accepted target

The redesigned evaporator has no qualified charge mass yet. Neither the donor's 15/23 g nor an automatic volume overage is the final recipe.

2 Keep the scale honest

The owned scale resolves 0.1 g. Check its response with a known mass; support hoses independently so their pull cannot change the reading.

3 Record net delivery

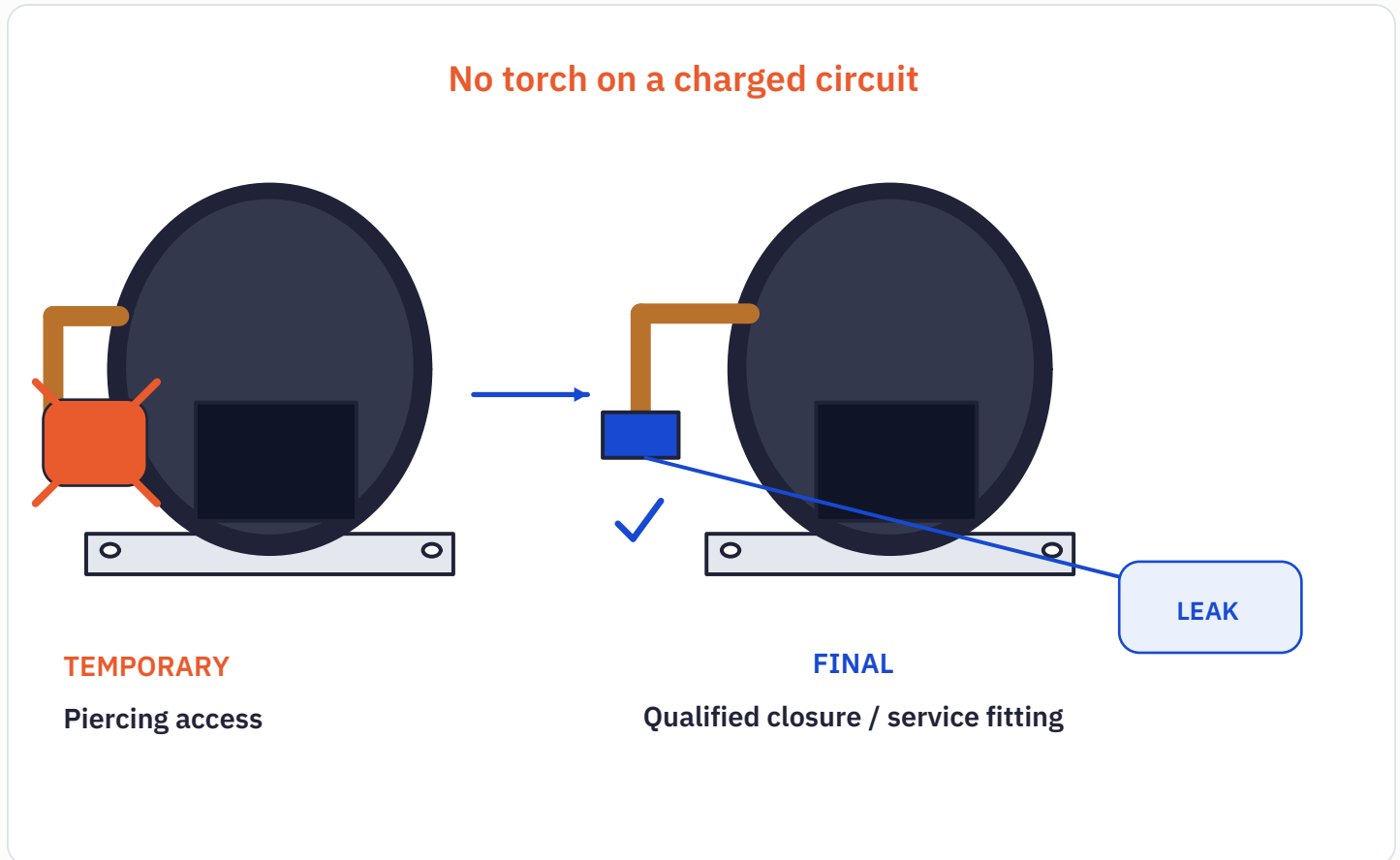
Follow the technician's refrigerant state/orientation and valve sequence. Record start/end mass, hose inventory and net to circuit. The project metering target is +/-1 g.

HOLD

Charging waits for the calibrated target, refrigerant grade and a repeatable net-delivery method.

Finish the process access

Leave a qualified permanent closure or service fitting, then check it after the charging connection is removed.



1 Use the specified closure

The technician's plan establishes the final fitting/closure and when it is made. A charged circuit must not go directly to a torch.

2 Remove temporary piercing access

Supco's FAQ warns solderless piercing valves can leak over time and says they should not remain after repair. Do not leave the BPV31 as the lifetime service point.

3 Check the final state

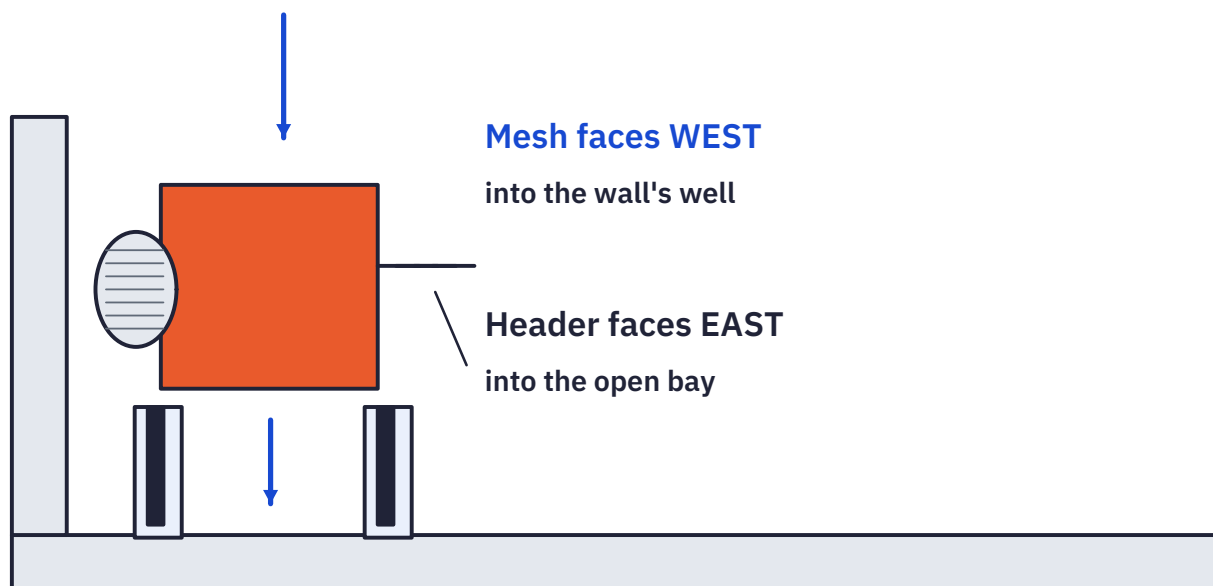
After disconnection, test the final closure, caps/threads and every joint with the qualified R600a leak method. Record detector identity and proof check.

DONE WHEN

The final closure is recorded and leak-tested; no temporary saddle is counted as the permanent seal.

Place the gas sensor first

Install the MQ-6 in the open floor strip before the compressor makes the slot hard to reach.



Install before the compressor blocks access.

Card is on edge; lower until it lands on its shoulder.

1 Orient the card on edge

The card runs along the west (-X) flank. Mesh faces west into the wall well; its header faces east into the bay.

2 Lower into the grooves

Drop the card straight between the two printed posts until it lands on their shoulder. Route its loom without lifting the card or blocking the low floor zone.

3 Check installed response later

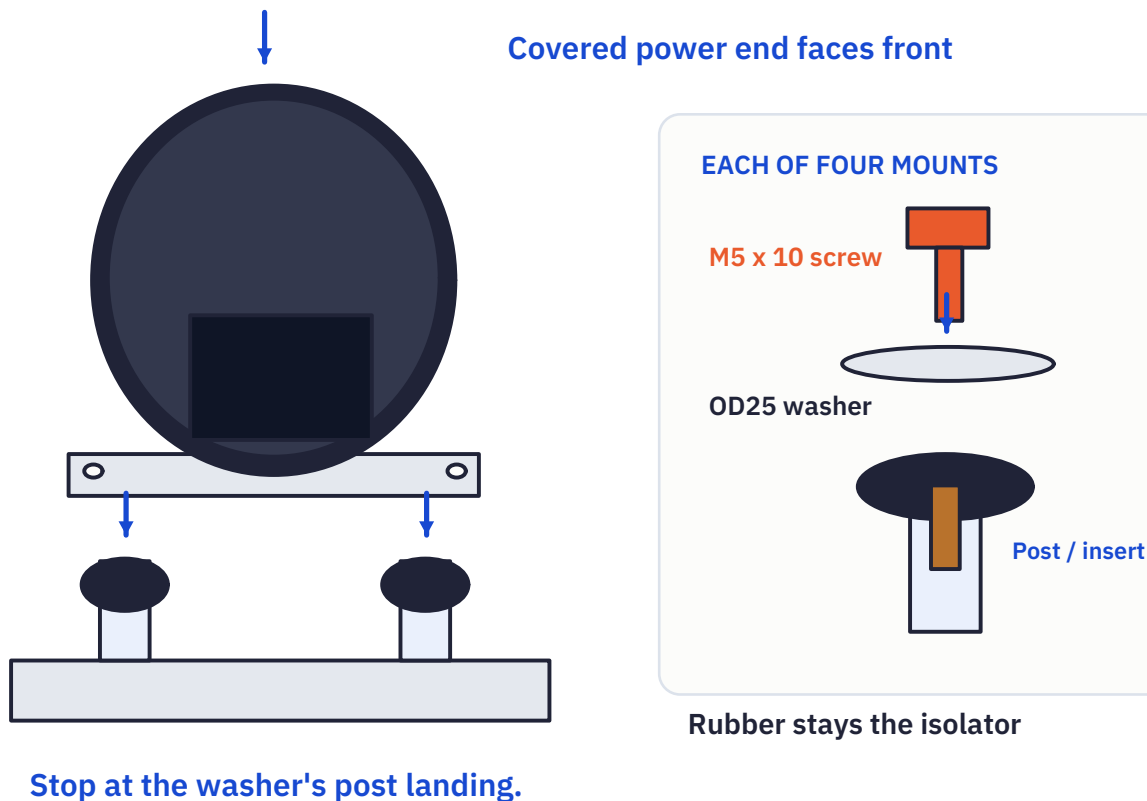
A located sensor is not a commissioned interlock. Prove module polarity, clean-air behavior and actual compressor shutdown under the approved test method.

CHECK

The gas module is reachable, correctly oriented and retained; its threshold/shutdown evidence remains open.

Lower the compressor

Mount the connected refrigeration assembly without transferring its weight into the copper lines.



1 Face the power end forward

Keep the original covered terminal/PTC assembly intact. Support the cold core and condenser while lowering the plate straight onto all four floor posts.

2 Use the four intended stacks

M5 x 10 SHCS with OD25 fender washers, into RX-M5x9.5 inserts. The posts rise through the rubber grommets; keep the grommets seated.

3 Stop at the post crown

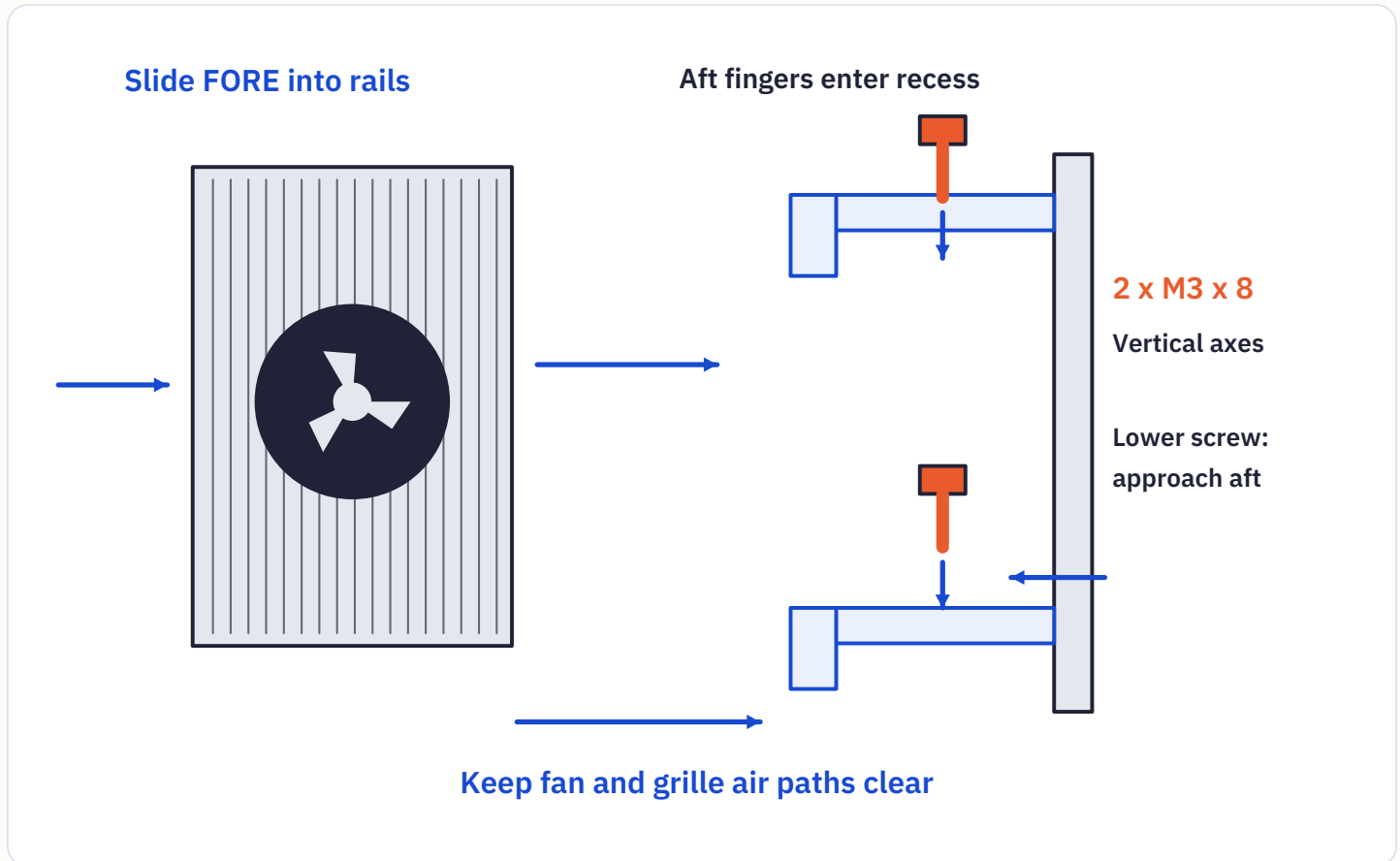
The washer is intended to land on the post after nominal 0.4 mm flange squeeze. Check the real bearing path and isolation; there is no accepted torque or load-life result.

DONE WHEN

All four stacks land without crushing rubber, hanging on tubework or forcing an insert.

Seat the condenser and fan

Use the printed rails and aft fingers to carry the condenser block above the slab.



1 Enter the aft recess

Feed the aft end onto the mounting fingers, then slide fore until the front flanges enter the rail grooves and the face meets their shoulder.

2 Close the two mounts

Drive the two M3 x 8 screws down through the donor's aft flange holes into the fingers' inserts. Approach the lower screw from aft with the bay still open.

3 Keep the air path open

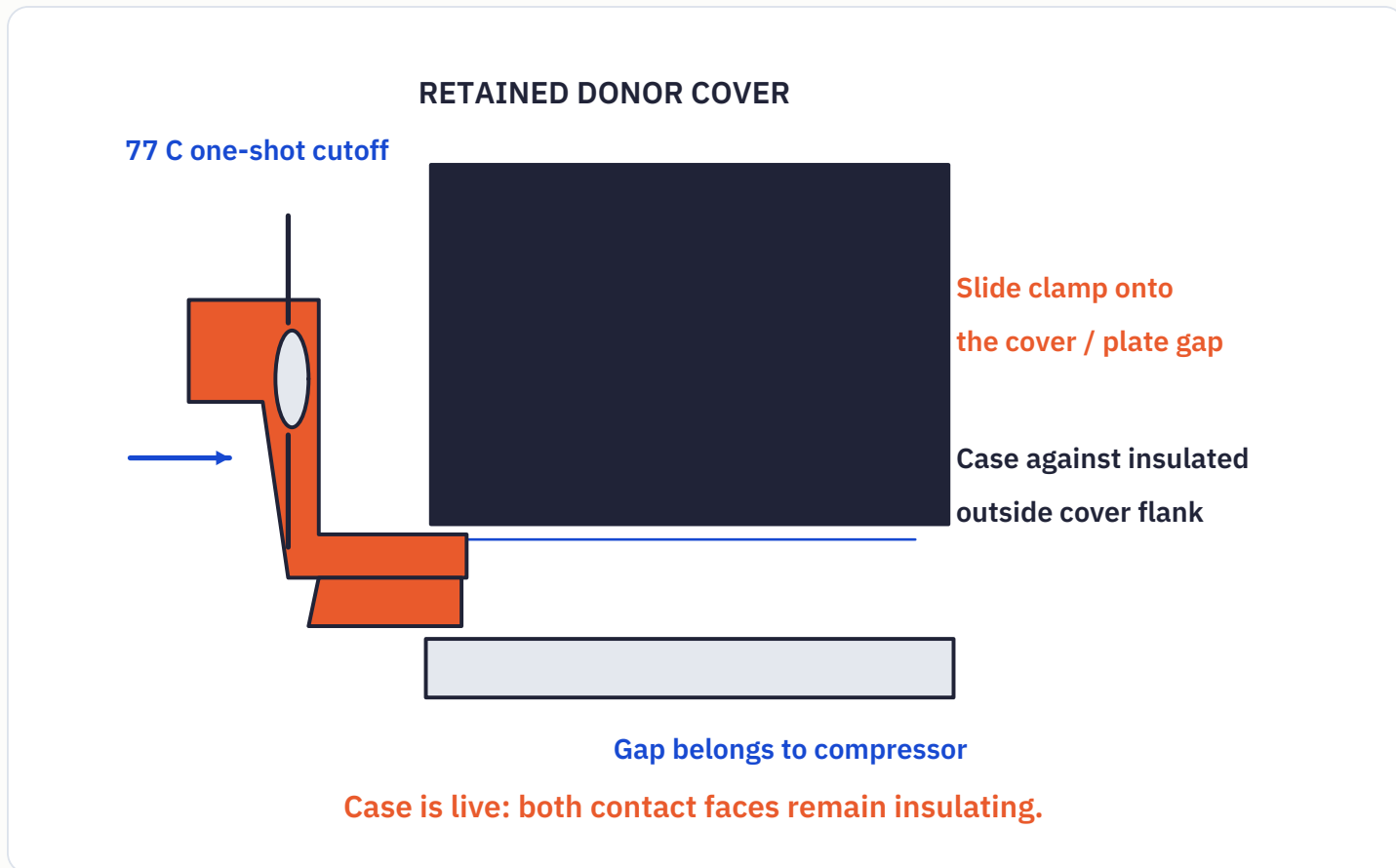
Confirm the native fan's installed flow direction, clear fins and cross-cabinet inlet/exhaust grilles. Inspect every attached line for stress or rubbing.

CHECK

The block is supported by its mounts, the fan turns freely and airflow is confirmed on the real donor.

Seat the thermal cutoff

The 77 C one-shot fuse reads through contact with the outside flank of the retained donor power cover.



1 Keep the live case insulated

The cover and printed clamp are the insulating contact surfaces. Keep bare case/leads clear of the compressor can and plate; wiring follows the AC schedule.

2 Place the case and clamp

Seat the fuse along its channel against the cover flank. Slide the clamp's leaves into the cover-to-plate gap so the clamp rides with the compressor.

3 Inspect the actual fit

The fuse must contact the cover without damage or a loose air gap. Nominal CAD diameter does not prove contact across case/print tolerance; preserve required straight lead length.

HOLD

Installed retention, contact and insulation need bench evidence before the cutoff is credited as protection.

Prove the first start

Finish electrical commissioning and leak checks before a brief compressor smoke test.

thermal

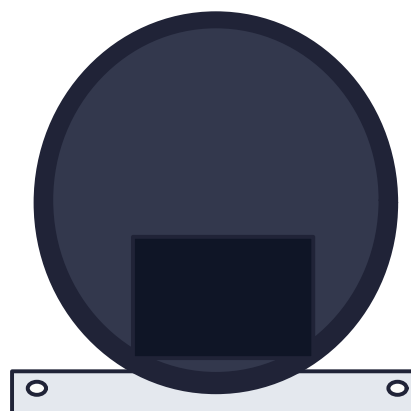
0x28 DS18B20 TANK

0x10 DS18S20 COIL

status

Probe health / gas clear

Relay + fan + off timer



Brief empty-core smoke test

30-60 s under a commissioned control

Observe / stop / honor restart guard

RECORD ACTUAL CURRENT

Compare with donor nameplate

1 Read real probes

Use thermal and status on the shipping firmware. Require family 0x28 on the tank and 0x10 on the coil; de-energized ambient checks are within +/-2 C.

2 Prove the control path

Use a commissioned bench control, with gas interlock and condenser fan working. Honor the 3-minute firmware off guard and any longer donor restart requirement.

3 Observe and stop

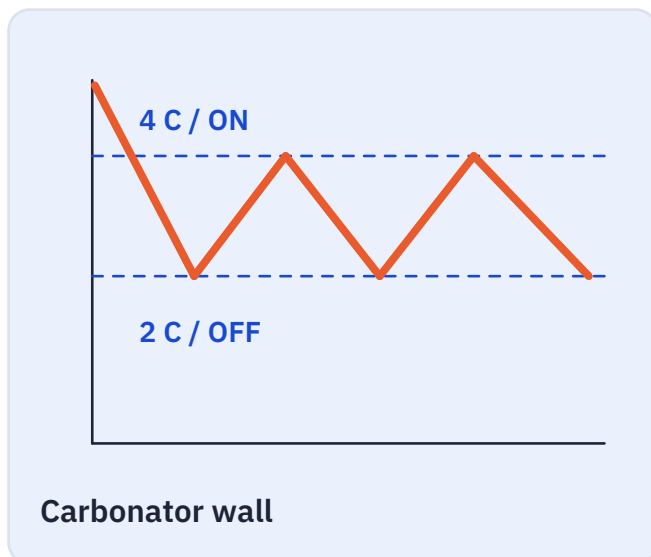
For the empty-core smoke test, 30-60 s is the procedure's brief window. Record actual nameplate-compatible current and falling coil temperature; stop for hum, overload or a leak.

CHECK

This proves switching and probe location only. It does not establish water-load performance or lifetime.

Prove the loaded cold loop

Commission refrigeration with the documented water load and verify the actual control limits.



COIL FREEZE GUARD

-8 C / trip

-5 C / recover

3 min minimum off

60 s minimum on

Stale probe / 30 s

Record the actual water-load test

Ambient / fill / charge / temperatures / cycles / current

1 Confirm current firmware values

Tank: on at 4 C, off at 2 C. Coil: trip at -8 C, recover at -5 C. Minimum off 3 min, on 60 s; stale probe after 30 s parks cooling.

2 Run the specified loaded test

Record water fill, ambient, actual charge, starting temperatures, pull-down, outlet-water temperature, coil trend, current and cycle timing.

3 Keep acceptance distinct

The acceptance document's 8-hour window and 10-70% duty band are proposed defaults. Missing loaded measurements and a qualified charge block production acceptance.

DONE WHEN

The logged control behavior and load result meet a committed test specification, with no leak alarm.

Repair through the full sequence

A leak or abnormal operation returns to diagnosis and controlled service, not direct reheating.

- 1 POWER OFF**
Isolate + identify
- ↓
- 2 REMOVE CHARGE**
Controlled HC route
- ↓
- 3 CLEAR / REPAIR**
Nitrogen + new drier
- ↓
- 4 TEST + EVACUATE**
Pressure + microns
- ↓
- 5 METER + CLOSE**
Qualified target

A charged or leaking circuit never goes directly to heat.

1 Stop and identify the fault

Isolate power, keep the area ventilated and use the approved leak/diagnostic method. A gas alarm, repeated freeze trip or overload is not a tune-by-guessing cue.

2 Remove charge under the right route

The completed dispenser's end-use basis and HC equipment govern removal. Do not borrow the donor's venting exemption.

3 Repeat the relevant proof

After opening: replace drier, clear/purge, repair, test, evacuate, meter charge and finish access. Record the cause and repeat the affected installed/load checks.

CHECK

Any route that skips charge removal or residual-fuel clearing before heat fails this sequence.

Keep the unit's evidence

The guide supplies order and pictures; the per-unit record establishes which work and measurements were completed.

PER-UNIT REFRIGERATION RECORD

☐ Donor / label / actual topology

Pages 2-3

☐ Coil / probes / routed protrusions

Pages 4-8

☐ HC setup / drier / qualified joints

Pages 10-17

☐ Micron trace / charge / final closure

Pages 18-20

☐ Mount / airflow / controls / load test

Pages 21-26

Name / serial / date / measurements / scope

1 Keep the measured result

Attach donor labels, coil/probe checks, service setup, joint qualifications, leak record, vacuum trace, net charge, final closure and installed control/load results.

2 Use the written sources

The footer names the responsible procedure or part reference. Click a source footer for an external service source where supplied; the README indexes local documents.

3 Keep this guide manual

Rebuild it deliberately when its operation changes. The PDFs and source receipt are committed artifacts; the appliance build does not redraw them.

HOLD

Open service limits, joints, charge and commissioning claims are visible at their operation, not signed off by this book.